Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 10

Lecture 19 Quantum Approximate Optimization Algorithm (QAOA)

19.1 Review: QUBO Problems

19.2 QAOA Algorithm

19.3 Example QAOA Algorithm

19.4 Other Variational Quantum Algorithms

Lecture 20 Portfolio Optimization using Quantum Computing

20.1 What is a financial portfolio?

20.2 Portfolio optimization problem

20.3 Portfolio optimization to QUBO

20.4 Solving portfolio optimization problem via QAOA

20.5 Two assets example

Practice Class

QAOA Algorithm

Recap: QUBO Problems

QUBO stands for "Quadratic Unconstrained Binary Optimization"

The cost function (which we want to minimize) is:

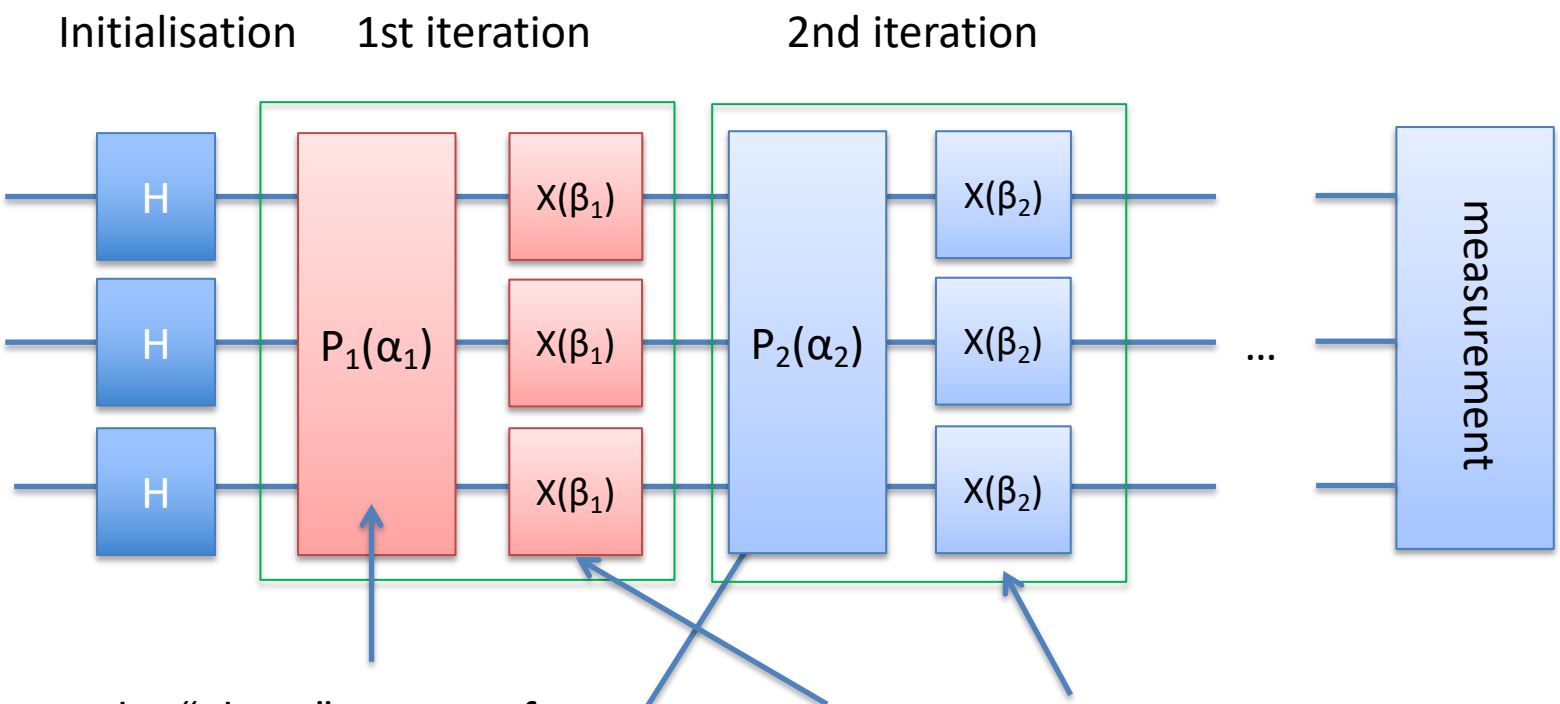
$$E(x_1, \ldots, x_n) = \sum_i c_i x_i + \sum_{i,j} Q_{ij} x_i x_j$$

Where x_i are Boolean (binary) variables, either 0 or 1.

Quadratic term

Recap: QAOA circuit

Overall the QAOA circuit looks like the following:



extract E value
re-set
parameters
α_i and β_i
repeat until min
is found

The "Phase" parts performs operations given by the structure of the quantum cost function

The X-parts performs X rotations on all qubits

This week we will build on this base to look at how finance problems can be solved on a quantum computer.

20.1 Financial Portfolios

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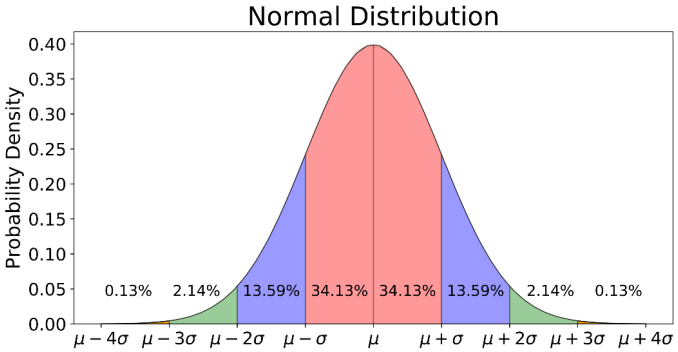
Lecture 20

Financial Portfolios

- Investment choices can be evaluated based on their risk-return characteristics.
  - Return is the % change in the invested amount in a given period (e.g., 1 year).
  - Risk is the variance (or standard deviation) of returns.
- Types of risk:
  - Systematic risk
    - Risk at the level of the whole market (e.g., economic recession).
  - Non-systematic risk.
    - Risk at the level of the individual assets (e.g., specific to a particular company).
- Investing in multiple assets, i.e., constructing a *portfolio*, can help in reducing non-systemic risk.



μ is the mean, σ is the standard deviation.

Two asset portfolio

- Consider a portfolio with two assets A and B with expected returns r_A and r_B, resp. The covariance matrix is as below ($\mathrm{Cov}(X, X) = \mathrm{Var}(\mathrm{X})$):

	Asset A	Asset B
Asset A	σ_A^2	σ_{AB}
Asset B	σ_{AB}	σ_B^2

Note: $\sigma_{AB} = \sigma_{BA}$

- Let the fraction of assets A and B in the portfolio be w_A and w_B, resp, where $w_A + w_B = 1$
- The expected return on the portfolio is:

$$w_A r_A + w_B r_B$$

- The expected variance of the portfolio is:

$$w_A^2\, \sigma_A^2 + w_B^2\, \sigma_B^2 + 2 w_A w_B\, \sigma_{AB}$$

Two asset portfolio: Example

- Consider a portfolio with two assets A and B each with expected returns 6%. The covariance matrix is as below:

	Asset A	Asset B
Asset A	0.4	−0.2
Asset B	−0.2	0.6

- Suppose we invest equal amounts in both A and B, i.e., $w_A = \mathrm{w_B} = 0.5$
- The expected return on the portfolio is:

$$w_A r_A + w_B r_B = 0.5 * 6 + 0.5 * 6 = 6\%$$

- The expected variance of the portfolio is:

$$w_A^2\ \sigma_A^2 + w_B^2\ \sigma_B^2 + 2 w_A w_B\ \sigma_{AB}$$

$$= (0.5)^2 * 0.4 + (0.5)^2 * 0.6 + 2 * 0.5 * 0.5 * (-0.2)$$
$$= 0.15$$

Question: Is it better to invest in both assets A and B or in only one asset?

Multi asset portfolio

- Consider a general case with n assets with expected returns $r_1, r_2, \dots, r_n$, resp, and covariance matrix:

	Asset 1	Asset 2	…	Asset n
Asset 1	σ_1^2	σ_{12}	…	σ_{1n}
Asset 2	σ_{21}	σ_2^2	…	σ_{2n}
…	…	…	…	…
Asset n	σ_{n1}	σ_{n2}	…	σ_n^2

Note:

1. $\sigma_{ij} = \sigma_{ji}$
2. $\sigma_{ii} = \sigma_i^2$ as $\mathrm{Cov}(X, X) = \mathrm{Var}(\mathrm{X})$

- Let the weights of each asset in the portfolio be $w_1, w_2, \dots, w_n$, resp, where $\sum_{i=1}^{n} w_i = 1$.
- Expected return of the portfolio: $\sum_{i=1}^{n} w_i r_i$
- Expected variance of the portfolio: $\sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$ where $\sigma_{ii} = \sigma_i^2$

20.2 Portfolio optimization problem

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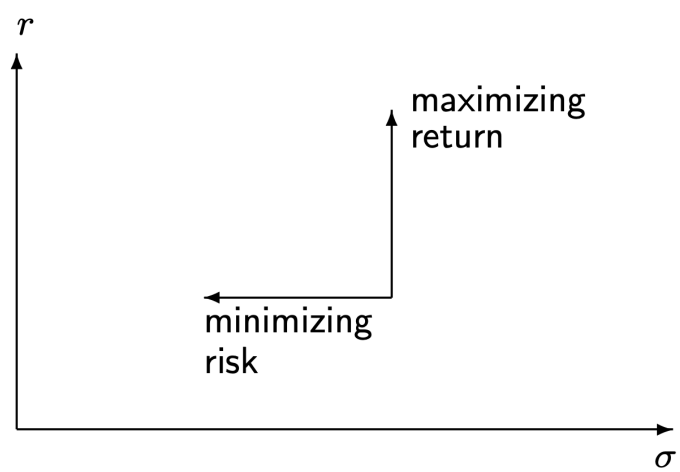
Lecture 20

Portfolio optimization problem

- How to select weights w_i of multiple assets in a portfolio?
- Let $\sum_{i=1}^{n} w_i r_i = \mathrm{r_P}$ be the portfolio return, $\sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij} = \sigma_P^2$ be the portfolio variance, and $\sum_{i=1}^{n} w_i = 1$.

- Three equivalent formalizations to determine asset weights w_i:
  - Maximize return given a level of risk σ^2:
    $\max r_P$ subject to $\sigma_P^2 = \sigma^2$
  - Minimize risk given the level of return r:
    $\min \sigma_P^2$ subject to $r_P = r$
  - Maximize return given a risk penalty λ:
    $\max r_P \; - \lambda \sigma_P^2$



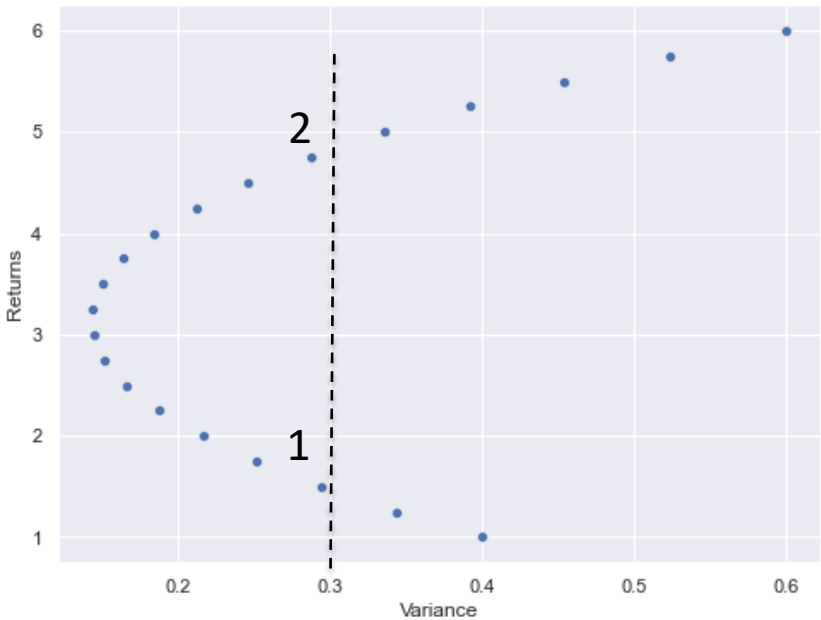
Note: Also commonly known as the mean-variance optimization problem.

Portfolio optimization problem: example

- Consider a portfolio with two assets A and B with returns 1% and 6% resp. The covariance matrix is as below:

	Asset A	Asset B
Asset A	0.4	-0.2
Asset B	-0.2	0.6

A portfolio required with a variance of 0.3



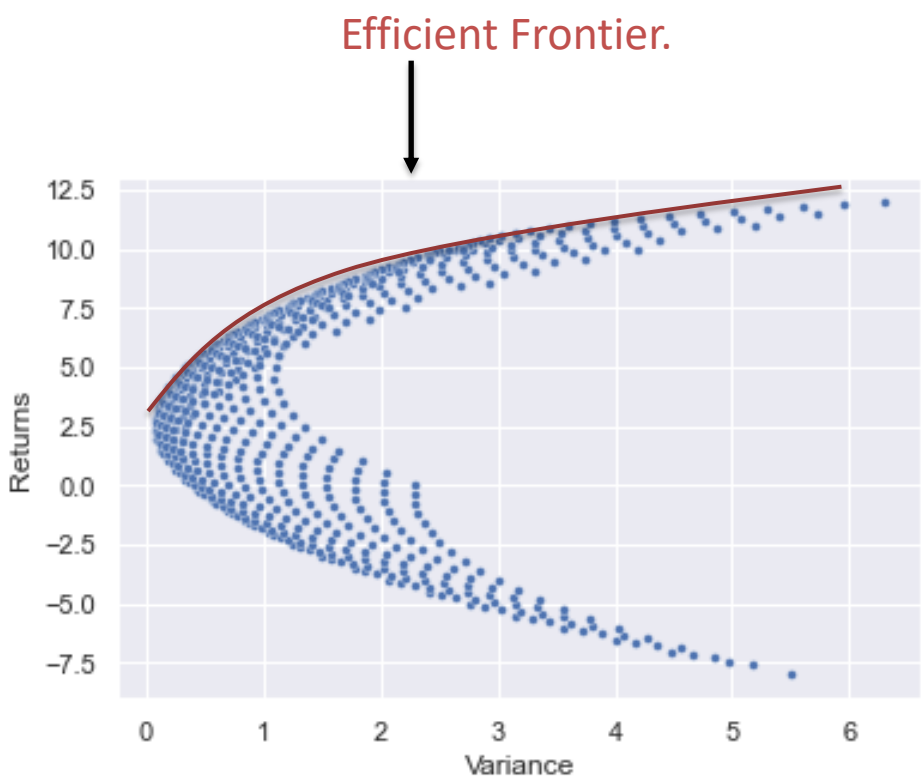
Question: Will you prefer portfolio 1 or portfolio 2?

Portfolio optimization problem: example

- Consider a portfolio with three assets A, B, and C with expected returns 1% ,5% and 2% resp.
  The covariance matrix is as below:

	Asset A	Asset B	Asset C
Asset A	0.4	-0.2	-0.1
Asset B	-0.2	0.6	0
Asset C	-0.1	0	0.3

- Efficient frontier: Set of optimal portfolios.



20.3 Portfolio optimization to QUBO

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Lecture 20

Portfolio optimization to QUBO

- Portfolio optimization (PO):

$$\max\left[\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}\right]$$

subject to $\sum_{i=1}^{n} w_i = 1$

This is the portfolio optimization problem with risk penalty.

- QUBO:

$$\min\left[\sum_{i=1}^{n} c_i x_i + \sum_{i=1}^{n}\sum_{j=1}^{n} Q_{ij} x_i x_j\right]$$

where x_i are binary (Boolean) variables.

Question: What are the differences in problems?

Portfolio optimization to QUBO

- Portfolio optimization (PO):

$$\max \left[\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}\right]$$

subject to $\sum_{i=1}^{n} w_i = 1$

- QUBO:

$$\min \left[\sum_{i=1}^{n} c_i x_i + \sum_{i=1}^{n}\sum_{j=1}^{n} Q_{ij} x_i x_j\right]$$

where x_i are binary (Boolean) variables.

1. PO is a maximization problem whereas QUBO is a minimization problem.
2. Variables (i.e., w_i) in PO can take real values whereas variables in QUBO (i.e., x_i) are binary.
3. PO has a constraint (weights should sum to 1) whereas QUBO is unconstrained.

Portfolio optimization to QUBO

1. PO is a maximization problem whereas QUBO is a minimization problem.

$$\max \left[\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}\right]$$

subject to $\sum_{i=1}^{n} w_i = 1$

Translate maximization to its equivalent minimization problem.

$$\min \; -1\left[\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}\right]$$

subject to $\sum_{i=1}^{n} w_i = 1$

$$\min \left[\lambda \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij} - \sum_{i=1}^{n} w_i r_i\right]$$

subject to $\sum_{i=1}^{n} w_i = 1$

Portfolio optimization to QUBO

2. Variables (i.e., w_i) in PO can take real values whereas variables in QUBO (i.e., x_i) are binary.

Translate weights into a dollar invested with a budget constraint.

Example: Consider two assets A and B where one decides to invest equally in each (i.e., $w_A = w_B = 0.5$).

	Asset A	Asset B	Total
Weights	0.5	0.5	1
Dollar amount	1	1	2

Portfolio optimization to QUBO

2. Variables (i.e., w_i) in PO can take real values whereas variables in QUBO (i.e., x_i) are binary. Translate weights into a dollar invested with a budget constraint.

Let the given budget be B. Transform weights $w_1, \dots, w_n$ to $x_{11}, \dots, x_{1B}, \dots, x_{n1}, \dots, x_{nB}$ binary variables where $x_{ij} = 1$ denotes a dollar j invested in asset i. For simplicity, we will denote $x_{11}, \dots, x_{1B}, \dots, x_{n1}, \dots, x_{nB}$ variables by $x_1, \dots, x_m$, where $m = nB$.

Example: Consider two assets A and B where one decides to invest equally in each (i.e., $w_A = w_B = 0.5$).

	x_1	x_2	x_3	x_4
Values	1	0	1	0
	Asset A		**Asset B**	

Note: Multiple equivalent assignments may exist.

Portfolio optimization to QUBO

2. Variables (i.e., w_i) in PO can take real values whereas variables in QUBO (i.e., x_i) are binary.

$$\min \quad [\lambda \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij} - \sum_{i=1}^{n} w_i r_i]$$

subject to $\sum_{i=1}^{n} w_i = 1$

Translate weights into a dollar invested with a budget constraint.

$$\min \quad [\lambda \sum_{i=1}^{m} \sum_{j=1}^{m} x_i x_j \sigma_{ij} - \sum_{i=1}^{m} x_i r_i]$$

subject to $\sum_{i=1}^{m} x_i = B$

Portfolio optimization to QUBO

3. PO has a constraint whereas QUBO is unconstrained.

Translate budget constraint into a penalty.

$$\min\ [\lambda \sum_{i=1}^{m}\sum_{j=1}^{m} x_i x_j \sigma_{ij} - \sum_{i=1}^{m} x_i r_i]$$

subject to $\sum_{i=1}^{m} x_i = B$

$$\downarrow$$

$$\min\ [\lambda \sum_{i=1}^{m}\sum_{j=1}^{m} x_i x_j \sigma_{ij} - \sum_{i=1}^{m} x_i r_i + \omega \left(\sum_{i=1}^{m} x_i - B\right)^2]$$

Where ω is a penalty for violating the budget constraint.

21.4 Solving portfolio optimization problem via QAOA

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Lecture 20

Portfolio optimization via QAOA

$$\min \; [\underbrace{\lambda \sum_{i=1}^{m}\sum_{j=1}^{m} x_i x_j \sigma_{ij}}_{\text{Risk term}} - \underbrace{\sum_{i=1}^{m} x_i r_i}_{\text{Return term}} + \underbrace{\omega \left(\sum_{i=1}^{m} x_i - B\right)^2}_{\text{Budget term}}]$$

1. Aggregate the quadratic and linear terms.
2. Transform x_i's to $z_i's$

Note: We will only highlight key concepts essential to the translation.

Portfolio optimization via QAOA: Risk term

QAOA: $H = \sum_{\boxed{i \neq j}} J_{ij} Z_i Z_j + \sum_i B_i Z_i$

$$\lambda \sum_{i=1}^{m} \sum_{j=1}^{m} x_i x_j \sigma_{ij}$$

Example:

$\lambda[x_1^2 \sigma_1^2 + x_1 x_2 \sigma_{12} + x_2 x_1 \sigma_{21} + x_2^2 \sigma_2^2]$

Question: How to deal with $i = j$?

Portfolio optimization via QAOA: Risk term

QAOA: $H = \sum_{\boxed{i \neq j}} J_{ij} Z_i Z_j + \sum_i B_i Z_i$

$$\lambda \sum_{i=1}^{m} \sum_{j=1}^{m} x_i x_j \sigma_{ij} = \lambda \sum_{i \neq j}^{m} x_i x_j \sigma_{ij} + \lambda \sum_{i}^{m} x_i \sigma_i^2$$

Example:

$\lambda[x_1^2 \sigma_1^2 + x_1 x_2 \sigma_{12} + x_2 x_1 \sigma_{21} + x_2^2 \sigma_2^2] = \lambda[x_1 x_2 \sigma_{12} + x_2 x_1 \sigma_{21}] + \lambda[x_1 \sigma_1^2 + x_2 \sigma_2^2]$

Question: How to deal with $i = j$?

Answer: Since x_i are binary, treat x_i^2 as linear terms.

x_i	x_i^2
0	0
1	1

Portfolio optimization via QAOA

	Quadratic	Linear	Constant
Risk term	$\lambda \sum_{i \neq j}^{m} x_i x_j \sigma_{ij}$	$\lambda \sum_{i}^{m} x_i \sigma_i^2$	
Return term		$-\sum_{i=1}^{m} x_i r_i$	
Budget term	$\omega \sum_{i \neq j}^{m} x_i x_j$	$\omega \sum_{i=1}^{m} x_i (1-2B)$	ωB^2

Constant can be ignored.

$$= \sum_{i \neq j}^{m} x_i x_j (\omega + \lambda \sigma_{ij}) + \sum_{i=1}^{m} x_i [\omega(1-2B) + \lambda \sigma_i^2 - r_i] + \omega B^2$$

QAOA: $H = \sum_{i \neq j} J_{ij} Z_i Z_j + \sum_{i} B_i Z_i$

Recap: Binary to energy

Typically when we write such energy functions we write in terms of the z variables:

$$z_i$$

(lower case z)

But the binary variables in terms of 0 or 1:

$$x_i = 0 \quad \text{or} \quad x_i = 1$$

Can convert between x_i and z_i using:

$$x_i \to \frac{z_i + 1}{2}$$

Recap: Solving via QAOA approach

QUI/Quantum Computer

(1) Prepare a trial state $|\psi(\alpha, \beta)\rangle$

on the quantum computer, where α, β represent parameters specifying k iterations of phase and X rotations.

(2) **Measure** to obtain the average energy, E.

Repeat these steps until the value of the energy converges

(3) Use a **classical optimization** technique such as the Nelder–Meade, determine new values of α, β that decrease E.

Classical

Lowest energy state → Solution to the initial problem

QAOA output to portfolio weights

How to translate a lowest energy state to asset weights of the optimal portfolio?

QAOA tells us that a state with the lowest energy is $|s_1 s_2 \dots s_m\rangle$.

s_i	0	1
z_i	1	-1
x_i	1	0

$$|s_1 s_2 \dots s_m\rangle$$

$$\downarrow \quad |0\rangle: z_i = 1, \quad |1\rangle: z_i = -1$$

$$z_1, z_2, \dots, z_m$$

$$\downarrow \quad x_i = \frac{z_i + 1}{2}$$

$$x_1, \dots, x_m$$

$\downarrow$ Each asset as B number of x variables.

$$x_{11}, \dots, x_{1B}, \dots, x_{n1}, \dots, x_{nB}$$

$$\downarrow \quad w_i = \frac{1}{B}\sum_{j=1}^{B} x_{ij}$$

$$w_1 w_2 \dots w_n$$

Tip: The values of s_i and x_i get flipped in the translation (e.g., $s_i = 0 \Rightarrow z_i = 1 \Rightarrow x_i = 1$).

21.5 Two asset example

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Lecture 20

Portfolio optimization problem: example

- Consider a portfolio with two assets A and B with returns 1% and 6% resp. The risk penalty is 20. The covariance matrix is as below:

	Asset A	Asset B
Asset A	0.4	-0.2
Asset B	-0.2	0.6

- Let the budget be \$2 and the budget penalty be 5. Corresponding x_i's are as below:

x_1	x_2	x_3	x_4
Variables for Asset A		Variables for Asset B	

Equivalent solutions

Violate budget constraint

- Example allocations:

x_1	x_2	x_3	x_4
1	0	0	1
\$1 in Asset A		\$1 in Asset B	

x_1	x_2	x_3	x_4
0	1	0	1
\$1 in Asset A		\$1 in Asset B	

x_1	x_2	x_3	x_4
1	0	0	0
\$1 in Asset A		\$0 in Asset B	

x_1	x_2	x_3	x_4
1	1	0	1
\$2 in Asset A		\$1 in Asset B	

Portfolio optimization problem: example

$$\min \ [\lambda \sum_{i=1}^{m} \sum_{j=1}^{m} x_i x_j \sigma_{ij} - \sum_{i=1}^{m} x_i r_i + \omega \left(\sum_{i=1}^{m} x_i - B \right)^2]$$

x_i: x_1, x_2, x_3, x_4

λ: 20

r_i: $r_1 = r_2 = 1, r_3 = r_4 = 6$

σ_{ij}: see table

ω: 5

B: 2

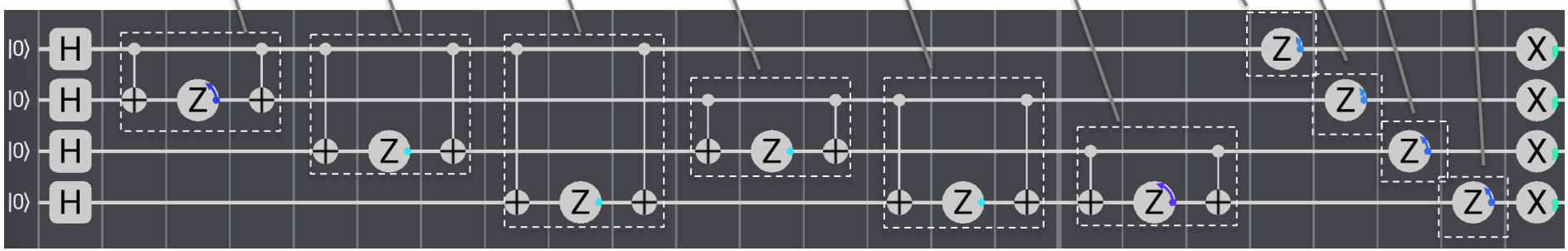
	1 (A)	2 (A)	3 (B)	4 (B)
1 (A)	0.4	0.4	-0.2	-0.2
2 (A)	0.4	0.4	-0.2	-0.2
3 (B)	-0.2	-0.2	0.6	0.6
4 (B)	-0.2	-0.2	0.6	0.6

$$26x_1x_2 + 2x_1x_3 + 2x_1x_4 + 2x_2x_3 + 2x_2x_4 + 34x_3x_4 - 8x_1 - 8x_2 - 9x_3 - 9x_4 + 20.0$$

Portfolio optimization problem: example

$$26x_1x_2 + 2x_1x_3 + 2x_1x_4 + 2x_2x_3 + 2x_2x_4 + 34x_3x_4 - 8x_1 - 8x_2 - 9x_3 - 9x_4 + 20$$

$$6.5z_1z_2 + 0.5z_1z_3 + 0.5z_1z_4 + 0.5z_2z_3 + 0.5z_2z_4 + 8.5z_3z_4 + 3.5z_1 + 3.5z_2 + 5.0z_3 + 5.0z_4$$



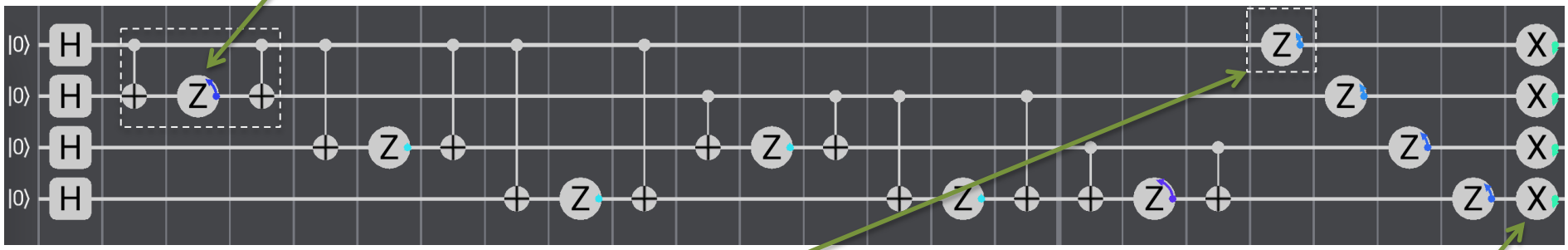
We need to input the angles α and β

Portfolio optimization problem: example

A deeper look at the quantum algorithm.

$$6.5z_1z_2 + 0.5z_1z_3 + 0.5z_1z_4 + 0.5z_2z_3 + 0.5z_2z_4 + 8.5z_3z_4 + 3.5z_1 + 3.5z_2 + 5.0z_3 + 5.0z_4$$

1. What should be the angle of rotation here?



Let $\alpha = 0.2$ and $\beta = 0.1$

2. What should be the angle of rotation here?

3. What should be the angle of rotation here?

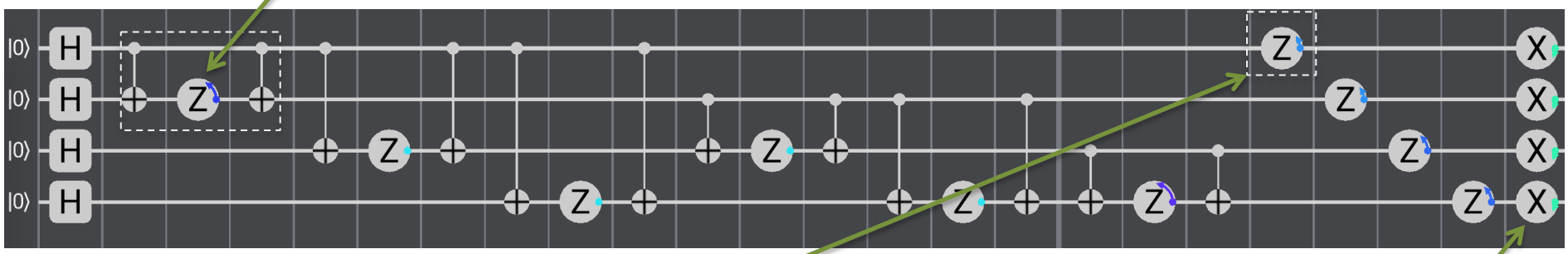
Note: You should create global parameters α and β in QUI.

Portfolio optimization problem: example

A deeper look at the quantum algorithm.

$$6.5z_1z_2 + 0.5z_1z_3 + 0.5z_1z_4 + 0.5z_2z_3 + 0.5z_2z_4 + 8.5z_3z_4 + 3.5z_1 + 3.5z_2 + 5.0z_3 + 5.0z_4$$

$$-\frac{6.5 * \alpha}{2} = -0.65$$



Let $\alpha = 0.2$ and $\beta = 0.1$

$$-\frac{3.5 * \alpha}{2} = -0.35$$

$$\beta = 0.1$$

Note: You should create global parameters α and β in QUI.

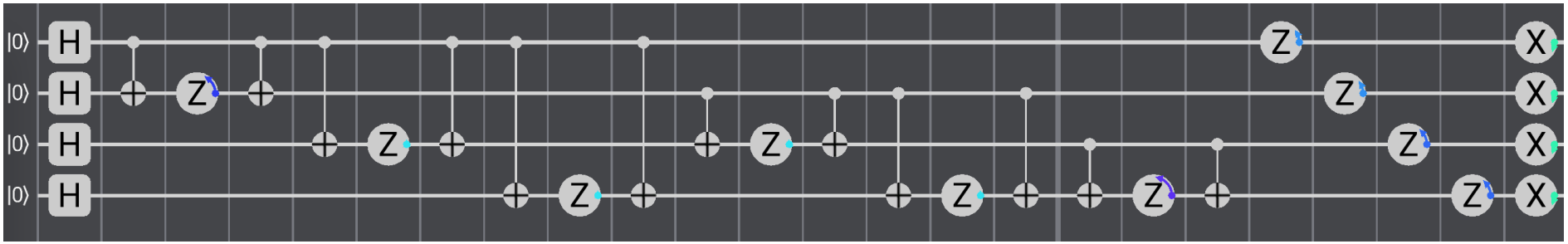
Portfolio optimization problem: example

$$26x_1x_2 + 2x_1x_3 + 2x_1x_4 + 2x_2x_3 + 2x_2x_4 + 34x_3x_4 - 8x_1 - 8x_2 - 9x_3 - 9x_4 + 20$$

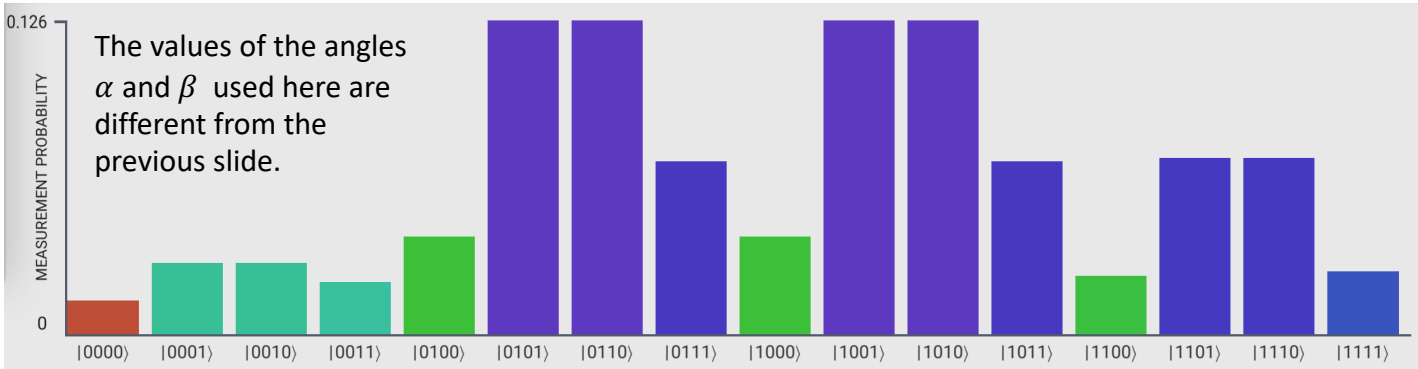
$$\downarrow$$

$$6.5z_1z_2 + 0.5z_1z_3 + 0.5z_1z_4 + 0.5z_2z_3 + 0.5z_2z_4 + 8.5z_3z_4 + 3.5z_1 + 3.5z_2 + 5.0z_3 + 5.0z_4$$

$\downarrow$ ← We need to input the angles α and β

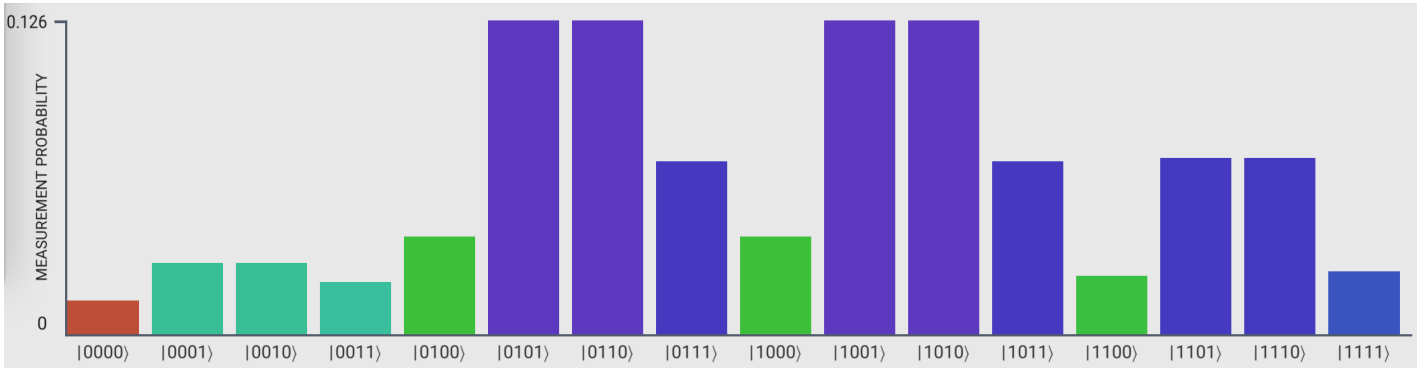


$\downarrow$



The values of the angles α and β used here are different from the previous slide.

QAOA output to portfolio weights



With the optimal values of α and β we should get states with lowest energy (highest probability in QUI). To compute the weights in the original problem we have to compute the dollar amount invested in each asset (based on the lowest energy state). We get multiple equivalent optimal solutions.

State	$x_1x_2x_3x_4$	Invested amounts $(x_1 + x_2, x_3 + x_4)$	w_1, w_2
$\|0101\rangle$	1010	1, 1	0.5,0.5
$\|0110\rangle$	1001	1,1	0.5,0.5
$\|1001\rangle$	0110	1,1	0.5,0.5
$\|1010\rangle$	0101	1,1	0.5,0.5

Week 10

Lecture 19 Quantum Approximate Optimization Algorithm (QAOA)

19.1 Review: QUBO Problems

19.2 QAOA Algorithm

19.3 Example QAOA Algorithm

19.4 Other Variational Quantum Algorithms

Lecture 20 Portfolio Optimization using Quantum Computing

20.1 What is a financial portfolio?

20.2 Portfolio optimization problem

20.3 Portfolio optimization to QUBO

20.4 Solving portfolio optimization problem via QAOA

20.5 Two assets example

Practice Class

QAOA Algorithm

Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8